The NORMAN Suspect List Exchange (NORMAN-SLE) in 2025: Global Reach and FAIR Data growth

Hiba Mohammed Taha[1], XXXX, Emma L. Schymanski[1]*

[1] Luxembourg Centre for Systems Biomedicine (LCSB), University of Luxembourg, 6 avenue du Swing, 4367, Belvaux, Luxembourg.

ORCIDs HMT: 0000-0001-7820-4335, XXX, ELS: 0000-0001-6868-8145.

* Correspondence: ELS: emma.schymanski@uni.lu.

Abstract

The NORMAN Suspect List Exchange (NORMAN-SLE) continues to evolve as a central, FAIR (Findable, Accessible, Interoperable, Reusable) resource for the global environmental scientific community. Established by the NORMAN Association, the NORMAN-SLE has now been supporting suspect screening efforts with curated chemical information and expert knowledge for over a decade. Since May 2022, the NORMAN-SLE has welcomed 34 new suspect lists and 25 updated versions of pre-existing lists, including 16,502 new compounds. As of September 2025, the NORMAN-SLE includes 133 separate suspect list collections from over 100 global contributors, totaling 133,573 unique compounds. Between 2022 and 2025, the platform recorded a three to five-fold increase in Zenodo views, downloads, and citations, reflecting growing scientific adoption and visibility. The collection now includes new chemical classes such as Ionic Liquids, Liquid Crystal Monomers (LCM), Substances of Unknown or Variable composition, Complex reaction products or Biological materials (UVCBs), Endocrine Disrupting Chemicals (EDCs), and Food Contact Chemicals (FCCs), alongside expanded per- and polyfluoroalkyl substances (PFAS) and transformation product (TPs) coverage. Data curation activities corrected misidentified entries, generated new PubChem Compound identifiers, and improved cross-platform interoperability. Coupled with the continued assignment of Digital Object Identifiers (DOIs), implementation of traceable versioning and facilitation of expanded access through the Zenodo platform and the NORMAN Network infrastructure, NORMAN-SLE advances the principles of Open Science. Future directions will explore creating a relational database for NORMAN-SLE and a user-oriented web interface to improve data access and exploration. The continued growth of the NORMAN-SLE demonstrates how open, FAIR chemical data can support scientific studies, catalyze collaboration, and promote further research initiatives.

Keywords: Suspect screening, High resolution mass spectrometry, Non-target screening, Open science, FAIR (Findable Accessible Interoperable Reusable) data, Data exchange, Cheminformatics, Exposomics, Environmental contaminants, Chemicals of emerging concern

Background

The increasing numbers of Chemicals of Emerging Concern (CECs) in the environment such as pesticides, transformation products (TPs), persistent “forever chemicals” like per- and polyfluoroalkyl substances

(PFAS) and chemicals associated with plastics pose a wake-up call for the urgency to safeguard human and environmental health [1]. Since targeted monitoring approaches cannot capture all compounds detected in complex environmental samples, suspect screening coupled with high resolution mass spectrometry (HRMS) serves as an efficient approach to search for the presence of suspected chemicals in environmental samples [2].

Suspect lists are predefined lists of chemicals of interest and/or suspected to occur in a given matrix. They enable more efficient screening of complex samples using the exact mass of compounds to find features of interest in a sample, and can also be used to create inclusion lists in advance of analysis. Several pieces of information are used to denote entries in the suspect list, including structural identifiers like the International Union of Pure and Applied Chemistry (IUPAC) International Chemical Identifier (InChI) [3], the hashed InChIKey [4], and Simplified Molecular Input Line Entry System (SMILES) [5], as well as descriptive fields such as short or IUPAC names, and/or database-specific identifiers like Chemical Abstracts Service (CAS number) [6], DSSTox substance identifier (DTXSID) [7] and PubChem Compound identifier (CID) [8]. Suspect lists also often contain expert curated information that bridges the gap between routinely monitored substances and emerging contaminants that are still overlooked. For instance, while regulated pesticides may appear within acceptable limits during monitoring, related transformation products that potentially contribute to the risk may remain undetected [9]. Consequently, suspect screening represents a powerful approach for prioritizing chemicals of concern and supporting regulatory decision-making. While non-target screening offers the potential for identifying unknown compounds, the speed and focused prioritization of suspect screening makes it a powerful strategy for systematic, large-scale environmental surveillance [2].

Environmental monitoring increasingly depends on access to accurate, open, and interoperable chemical information to make such approaches effective. Additionally, the growing complexity of chemical exposures in the environment and the need for robust environmental risk assessments has amplified the demand for FAIR (Findable, Accessible, Interoperable, Reusable) chemical data [10] [11]. Open and standardized chemical information supports regulatory compliance and enables researchers to efficiently integrate data across studies and platforms, enhancing both scientific discovery and public health protection. Challenges in chemical data discoverability remain due to fragmented sources, inconsistent identifiers, and varying curation standards. Community-driven curation helps address these issues by harmonizing datasets, standardizing identifiers, and incorporating expert knowledge into open resources. Such collaborative efforts are essential to ensure that chemical data are not only available but also usable across research, monitoring, and regulatory contexts [12].

The NORMAN Suspect List Exchange (NORMAN-SLE), established under the NORMAN Network, addresses this need by hosting community-contributed suspect lists to support HRMS-based screening efforts. Since its inception, the NORMAN-SLE has functioned as both a data repository and a collaborative platform, enabling harmonized data sharing across laboratories and regulatory bodies. The first NORMAN-SLE publication gives a comprehensive overview of the structure, FAIR implementation, and integration within the NORMAN Database System [13]. At that stage, 99 curated lists were available, encompassing more than 115,000 unique compounds linked to the NORMAN Substance Database (SusDat), PubChem [8] and the United States Environmental Protection Agency's (US EPA) CompTox Chemicals Dashboard [7]. These lists covered major contaminant classes including PFAS, transformation products, pharmaceuticals, pesticides, and regulatory monitoring lists. Over the subsequent three years, the NORMAN-SLE has undergone sustained growth in data volume, contributor participation, and public engagement.

This paper presents updates to the NORMAN-SLE since the original publication in May 2022 [13], focusing on quantifiable indicators of impact: data expansion, usage metrics, curation outcomes, and interoperability with global chemical databases. By examining these dimensions, the NORMAN-SLE's 2022-2025 evolution can be viewed as a model of how FAIR and Open Science principles bring meaningful benefits to science and society.

Methods

The NORMAN-SLE Workflow:

The methods behind the NORMAN-SLE are described in detail in the earlier publication [13]. Briefly, the starting point (blue box in Figure 1), are the suspect lists from various sources, which includes expert curated metadata, additional annotations and transformation information. FAIRifying these lists involves curation of the provided data and compilation of additional FAIR identifiers if missing, such as Name, InChI [3], InChIKey [4], SMILES [5], CAS number [6], CompTox [7] and PubChem [8] Identifiers. The compiled and curated lists are assigned unique codes and are then ready to be integrated into multiple platforms.

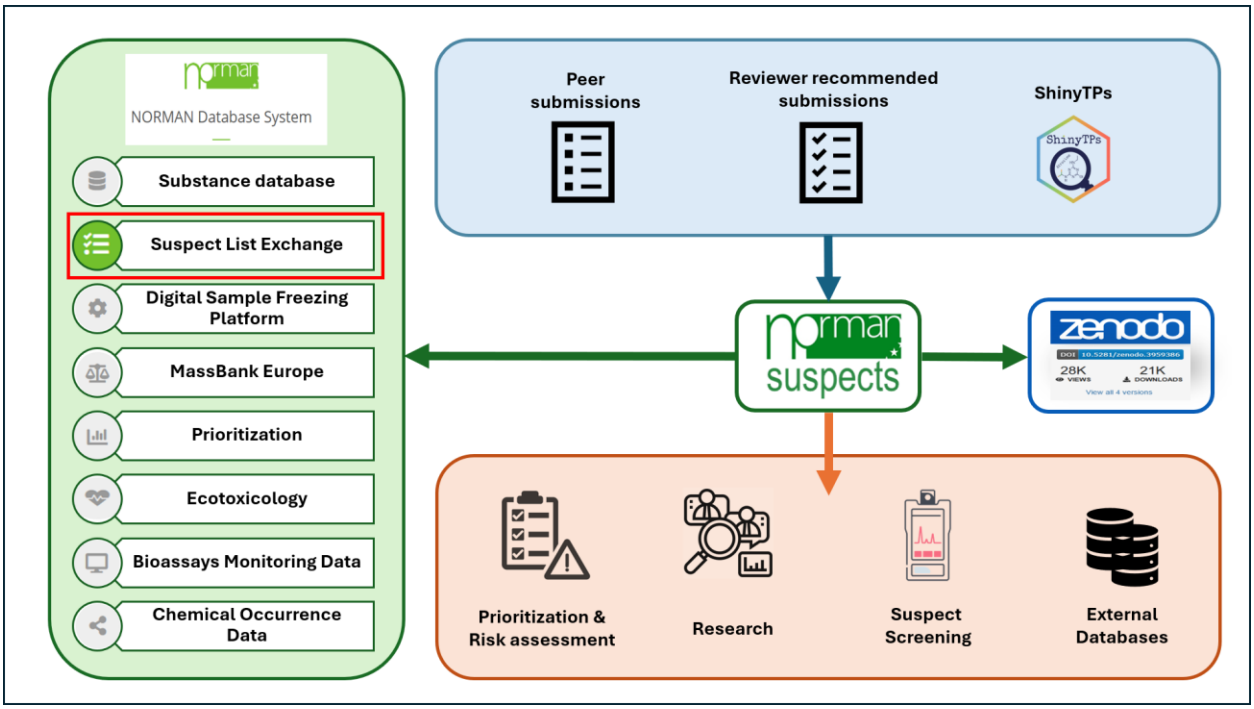

Figure 1: Overview of NORMAN-SLE Workflow.

The lists are first archived on the NORMAN-SLE community in Zenodo [14], yielding unique dataset DOIs that enables citation by users. This facilitates version management as well as providing statistics on views, downloads and citations. Further, the NORMAN-SLE content feeds back into the NORMAN Database System (NDS), forming the basis for the NORMAN Substance Database (NORMAN SusDat) that interconnects with the NORMAN Network's other activities (green box in Figure 1). The orange box illustrates the multiple applications that the NORMAN-SLE data supports further downstream in the workflow. This includes prioritization studies, risk assessment, advanced research, suspect screening and

integration in external databases like PubChem and CompTox. The automated workflow which harvests NORMAN-SLE content from Zenodo via mapping files and integrates it into PubChem [8] is described elsewhere [13].

Data Collection for Impact Metrics

To quantify the impact and use metrics of the NORMAN-SLE lists hosted on Zenodo, web scraping was performed using Python with the BeautifulSoup and Selenium libraries. The resulting workflow accessed the NORMAN-SLE community page on Zenodo and iteratively extracted the metadata of each list. Selenium was used to dynamically load webpage content and handle pagination, while BeautifulSoup parsed the HTML structure to retrieve relevant statistics. For each list, the total number of views, downloads, and citations was collected and then compiled into a structured dataset and saved as a dated csv file. The script was modified and rerun to extract the latest information and restructure the output data as required.

To assess the practical and scientific impact of NORMAN-SLE lists, two complementary Key Performance Indicators (KPIs) were considered: the Reuse Efficiency Score (RES) and the Recognition Index (RI). The RES was calculated as the ratio of total downloads to total views and quantifies how effectively data views translate into downloads. The RI was calculated as the ratio of total citations to total downloads, and reflects how frequently downloaded datasets contribute to visible research outputs.

Results and Discussion

Quantitative Growth and Community Expansion

Between 2022 and 2025, the NORMAN-SLE expanded from 99 to 133 suspect lists (S0 SUSDAT to S132 UJIGCAPCICCS), representing an increase of 18,325 unique compounds, with the total number now reaching 133,573 entries curated from around 100 contributors worldwide. The earlier publication [13] documented all lists up to S98, while this update introduces 34 new suspect lists (Table 1) and 25 updates to existing ones (major updates listed in Table 2). To ensure that the results presented here are internally consistent, the NORMAN-SLE activity was placed on hold during compilation of this manuscript (September and October 2025). All statistics and figures presented herein reflect the repository state as of this period. A comprehensive summary of all suspect lists, including their descriptions and number of entries, is provided as a CSV file in Additional File 1, while a snapshot of the NORMAN-SLE website as of September 2025 is included in Additional File 2.

Table 1: Summary of the newly added NORMAN-SLE datasets since 2022, with suspect list number (S), code, accompanying references and number of entries (lines in the file, in italics)

List	Code [Reference]	Entries	List	Code [(]Reference]	Entries
S99	ANSESEDC [15]	*906*	S100	PFASREACH [16]	*437*
S101	MTMDUST [17] [18]	*300*	S102	PARCPFAS [19]	*191*
S103	NORMANUVCB [20]	*232*	S104	UKVETMED [21]	*613*
S105	P65CHEM [22]	*991*	S106	AQUAlity19 [23]	*262*
S107	AQUAlity20 [24]	*387*	S108	SINLIST [25]	*1404*
S109	PARCEDC [26]	*7404*	S110	DUTCHUSE [27]	*1774*
S111	PMTPFAS [28]	*180*	S112	FCCMIGEX [29] [30]	*5063*

List	Code [Reference]	Entries	List	Code [(]Reference]	Entries
S113	SWISSPHARMA24 [31] [32]	*1814*	S114	SLUAMTPS [33] [34]	*3500*
S115	EIEQSDWLGC [35]	*119*	S116	REFCCS [36]	*205*
S117	PFASFCCDB [37] [38]	*140*	S118	PFASFCCMIGEX [39] [38]	*68*
S119	PLASTCHEM [40] [41]	*16542*	S120	DUSTCT2024 [42] [43]	*1210*
S121	EAWAGBBD [44] [45]	*1392*	S122	PBTPFASNTA [46] [47]	*1935*
S123	LITCHEMPLAST [48] [49]	*3500*	S124	IONICLIQUIDS [50]	*72*
S125	FLUOROIONS [51]	*67*	S126	SFCPFASFIONS [52]	*28*
S127	LCM [53] [54]	*1173*	S128	FLUOROLCM [55] [56]	*576*
S129	SURFPRO [57] [58]	*1624*	S130	NDSEXTRAS [59]	*243*
S131	NANJINGOPDE [60]	*101*	S132	UJIGCAPCICCS [61]	*250*

Table 2: Summary of the major updates in pre-existing NORMAN-SLE datasets since 2022, with suspect list number (S), code, accompanying references and number of entries (lines in the file) from 2022-2025.

List	Code [Reference]	Entries in 2022	Entries in 2025
S6	ITNANTIBIOTIC [62] [63]	676	5710
S68	HSDBTPS [64] [65]	101	963
S74	REFTPS [66]	146	550
S75	CYANOMETDB [67] [68]	2124	3080
S96	ECIPFAS [69]	258	319
S98	TIRECHEM [70] [71]	16	314

The contents of the NORMAN-SLE are shown in Figure 2, sorted by list size and colour coded according to whether the list remained static since 2022, was updated, or is new. Updates and new lists are reasonably evenly distributed by size. This representation highlights the growth of the NORMAN-SLE and the continuing engagement of a diverse community of contributors. The same overview sorted by list number, not size, and colour coded by list size is given in Supplementary Figure S1.

Beyond the overall expansion, some of the updated lists in the NORMAN-SLE saw major growth in the last three years, as shown in Table 2. The S6 ITNANTIBIOTIC list was extended to include a wider array of antibiotic compounds and their predicted metabolites, increasing its content to 5710 entries. The S68 HSDBTPS list expanded to 963 transformation products, reflecting the integration of additional data from Hazardous Substance Data Bank (HSDB) in PubChem via ShinyTPs [72]. The S74 REFTPS list was updated to include 404 new entries, incorporating newly reported transformation pathways from recent literature. Almost 1000 new entries were added to S75 CyanoMetDB, which now contains over 3080 cyanobacterial metabolites. The S96 ECIPFAS list broadened its coverage with additional PFAS, totaling 319 entries, while the S98 TIRECHEM list grew from being the smallest NORMAN-SLE list with tire-related chemicals to 314 entries, following the inclusion of new compounds of emerging concern from CompTox's TIRECRUMB list. The remaining updates to lists primarily involved minor technical revisions such as fixing SMILES or InChIKeys, adding new PubChem CIDs, merging or renaming columns, correcting metadata, and standardizing chemical information, without the addition of new compounds.

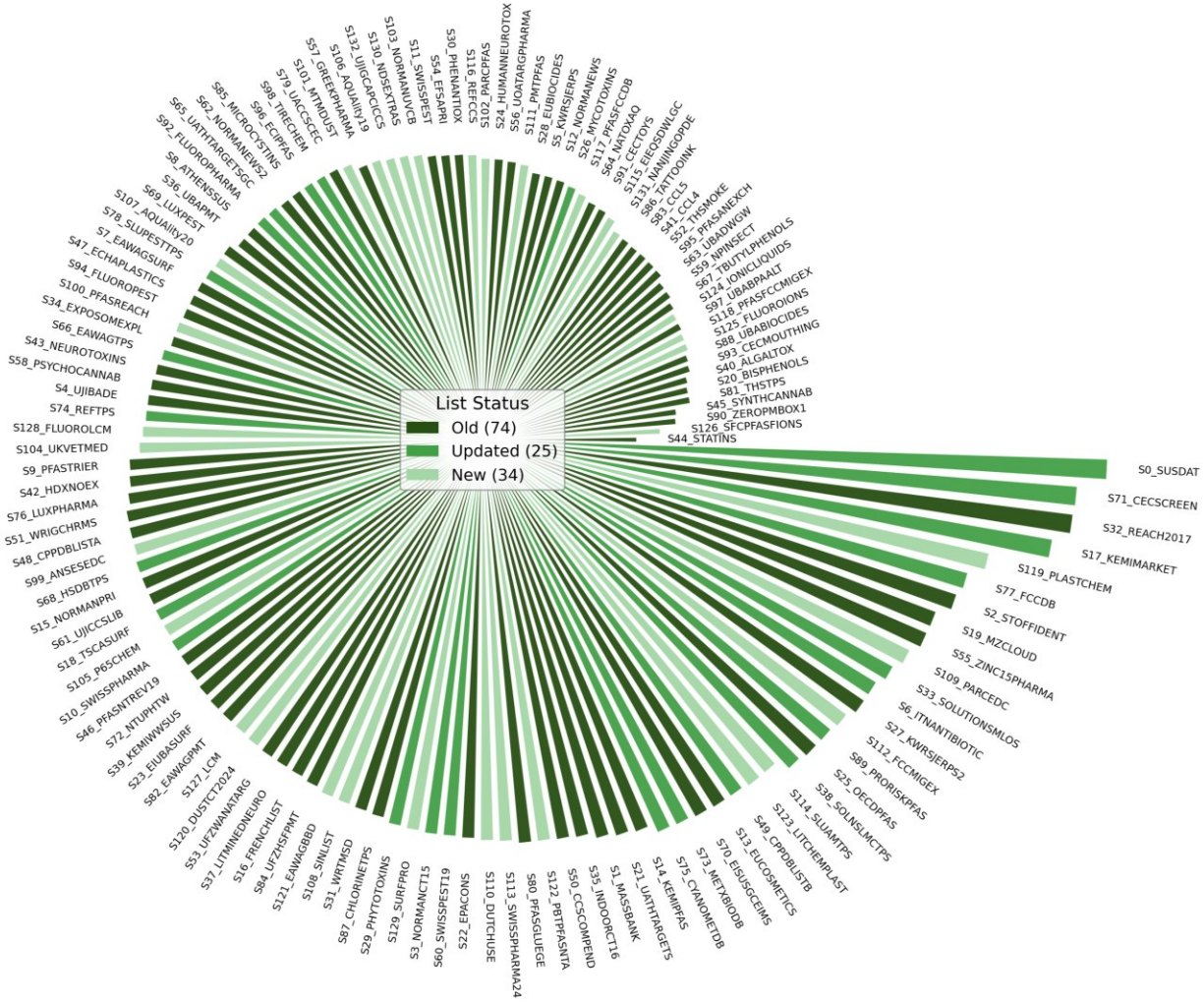

Figure 2: Contents of the NORMAN-SLE, sorted by list size and coloured by status (new, updated, static).

Diversification of Chemical Classes

A major outcome of the past three years was the increasing diversity in the coverage of chemical classes. This was achieved by adding both additional suspect lists and expanding metadata annotations of existing lists. Figure 4 shows the diverse nature of chemical classes added since 2022, including both well-known and emerging contaminant domains.

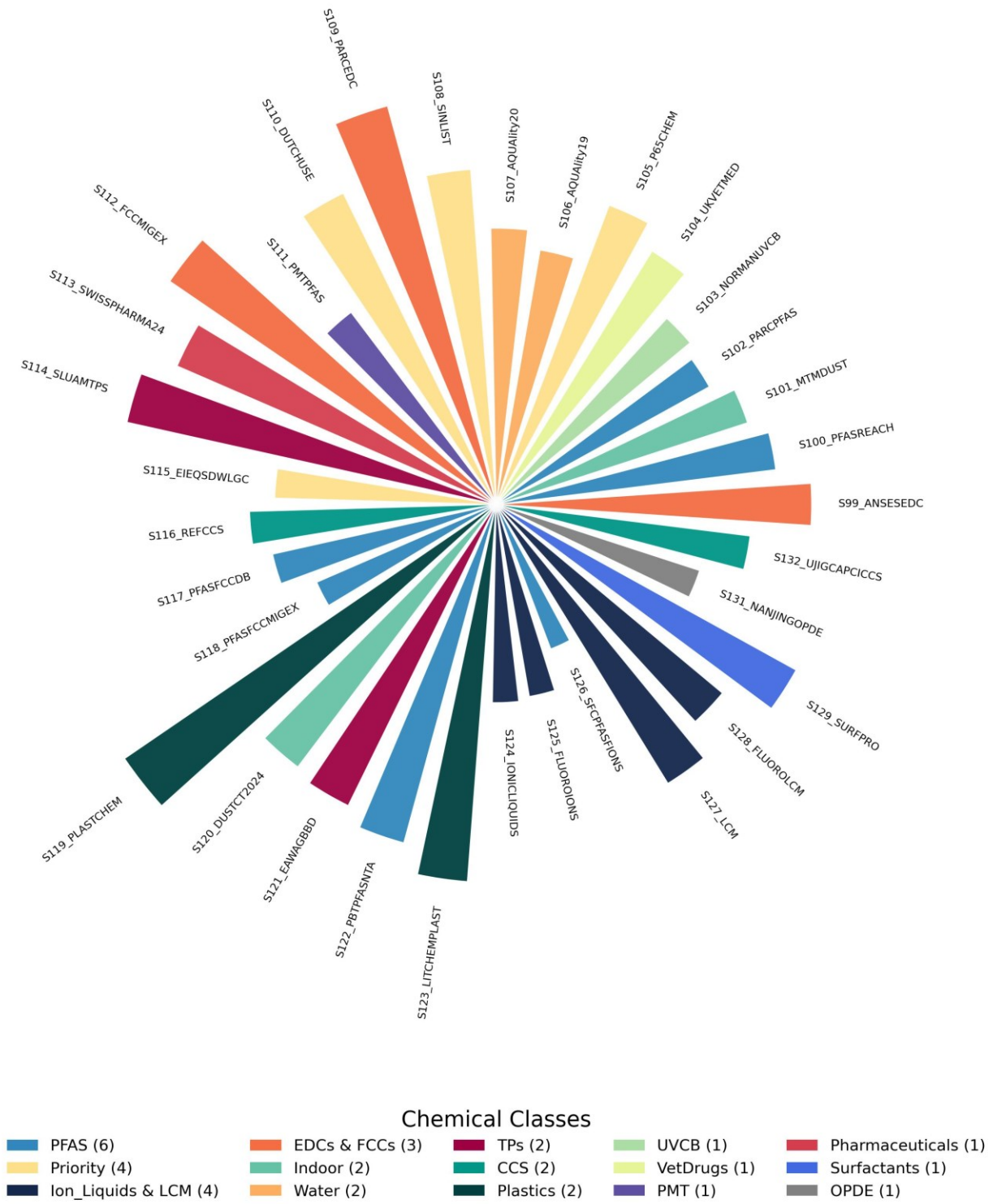

Figure 4: Chemical class coverage of newly added NORMAN-SLE Lists since 2022.

The PFAS category grew the most, with eight PFAS-related lists (S100_PFASREACH, S102_PARCPFAS, S111_PMTPFAS, S117_PFASFCC, S118_PFASFCCMIGEX, S122_PBTPFASNTA, S125_FLUOROIONS, and S126_SFCPFASIONS) added. This is primarily due to the continual increase in demand for information

about PFAS, but also reflects the need to integrate lists containing only PFAS (or only fluorinated compounds) into the *PFAS and fluorinated compound collections* section of the PubChem PFAS Tree [73]. This was the case for *e.g.* S117_PFASFCC and S118_PFASFCCMIGEX, which are the PFAS-containing subsets of S77_FCCDB and S112_FCCMIGEX. The increased attention to how consumers are exposed to FCCs is reflected in the fact that three new contributions were related to FCCs. Priority substances are captured through S105_P65CHEM, S108_SINLIST, S110_DUTCHUSE, and S115_EIEQSDWLGC, while medicinal compounds are represented by veterinary drugs (S104_UKVETMED) and pharmaceuticals (S6_ITNANTIBIOTIC). Environmental chemicals of interest for industrial or household applications include surfactants (S129_SURFPRO) and chemicals found in plastics (S119_PLASTCHEM, S123_LITCHEMPLAST), which add to the cross-sectoral coverage of SLE. There is also a growing representation of Transformation products (TPs) and related environmentally relevant substances. Lists such as S114_SLUAMTPS and S122_EAWAGBBD provide broad coverage of TPs that are often more persistent or mobile than their parent compounds. The PMT-relevant list S111_PMTPFAS complements this, while further additions expand the coverage of indoor (S101_MTMDUST, S120_DUSTCT2024) and water-related contaminants (S106_AQUAlity19, S107_AQUAlity20) as well as new experimental collision cross-section (CCS) values (S116_REFCCS, S132_UJIGCAPCICCS).

The NORMAN-SLE now also includes emerging and novel chemical classes. Endocrine-Disrupting Chemicals (EDCs) are represented through S99_ANSESEDC and S109_PARCEDC capturing compounds known or suspected to interfere with hormonal systems, whereas organophosphate diesters (OPDEs) are included via S131_NANJINGOPDE, covering degradation products of organophosphate flame retardants. Both standard and fluorinated ionic liquids are captured through S124_IONICLIQUIDS and S125_FLUOROIONS, and Liquid Crystal Monomers (LCMs) are represented by S127_LCM and S128_FLUOROLCM, targeting compounds of interest to electronic applications.

Collectively, these updates demonstrate a determined effort to expand both the depth and breadth of chemical class coverage in the NORMAN-SLE. By curating and openly sharing lists that are often underrepresented and integrating expert-curated metadata across these wide range of chemical classes, the repository now provides an enhanced platform for environmental monitoring and high-resolution mass spectrometry-based screening. The inclusion of emerging chemical classes such as LCMs, UVCBs, EDCs, and FCCs further highlights the diversification within SLE, displaying a shift in research focus towards new emerging contaminants. LCMs are important due to their widespread use in electronic materials and potential persistence in the various environmental matrices [74]. UVCBs on the other hand, pose challenges because of their complex compositions and regulatory blind spots [75]. EDCs [76] and FCCs [77] are also being widely researched due to their growing societal and regulatory concerns. These expansions also align with broader policy initiatives such as the EU Green Deal [78] and the Chemicals Strategy for Sustainability [79] which emphasize comprehensive chemical monitoring and safer chemical management. Thus, chemical diversification ensures that the NORMAN-SLE remains a functioning, community-driven resource capable of supporting both scientific research as well as regulatory decision-making.

Measuring FAIR Data Impact :

Between 2022 and 2025, NORMAN-SLE expanded notably from 99 to 133 lists, adding 18,325 unique compounds according to the PubChem NORMAN-SLE Tree [80] statistics (Table 3). The total live annotations listed on the PubChem NORMAN-SLE Source Page [81] nearly doubled, showing increased community engagement and curation activity. The largest list, S0_SUSDAT, a merging of all lists, continued to grow and remains central to the database.

While many chemical databases focus only on data volume, few assess how FAIR (Findable, Accessible, Interoperable, Reusable) principles translate into real-world visibility and reuse. To capture this, key use metrics were examined, namely total views, total downloads, and total citations from the NORMAN-SLE's Zenodo community over the three year period. While the total views reflect scientific discovery and user interest, total downloads indicate data reuse in workflows, while total citations demonstrate scientific trust and recognition. Usage statistics from the NORMAN-SLE Zenodo community compiled on 1 October 2025, also available on the NORMAN-SLE GitLab page, show an overall growing trend (Table 4). Total list views increased nearly five-fold (47,570 → 223,602), while downloads tripled (68,569 → 210,730), and citations rose from 40 to 274. This increased engagement aligns with FAIR data integration efforts and expanded international outreach. Detailed breakdown for all 133 lists, including total views, total downloads, and total citations, are provided in Additional file 3.

Table 3: Key summary statistics for the NORMAN-SLE 2022 v/s 2025.

Category	16 April 2022	01 October 2025
Total number of lists	99	133
Total unique compounds	115,248	133,573
Total live substances	117,071	139,400
Total live annotations	21,114	41,502
Largest list (# entries)	109,631 (*S0 SUSDAT*)	120,541 (*S0 SUSDAT*)
Smallest list (# entries)	18 (*S98_TIRECHEM*)	18 (*S44_STATINS*)

Table 4: Total Views, Downloads and Citations from 2022-2025 according to the NORMAN-SLE Zenodo Community [14].

Category	28 April 2022	04 April 2023	16 April 2024	01 October 2025
Total list views	47,570	86,005	143,734	223,602
Total list downloads	68,569	97,197	113,615	210,730
Total list citations	40	61	230	274

Among the top-performing lists (Figure 5), S13_EUCOSMETICS, S73_METXBIODB, and S0_SUSDAT consistently ranked highest in both views and downloads, reflecting strong relevance to regulatory and environmental research. New lists such as S109_PARCEDC and S100_PFASREACH quickly entered the Top 25, demonstrating keen interest in these new datasets, reflecting the current topics of EDCs, PFAS and chemicals in REACH.

No newly added lists have yet appeared in the Top 25 cited lists (Supplementary Figure S2), which is expected, given their recent release dates and the typical lag between usage and citation growth. This reflects both the gradual progression of reuse and recognition within open data ecosystems as well as the lag in time in publishing. This pattern suggests that a similar trend may also be observed in views and download statistics, highlighting the relevance of complementary KPIs, such as the Reuse Efficiency Score (RES) and the Recognition Index (RI) introduced above. In general, high RES value reflects strong reuse potential and user trust, while low RES values suggest high visibility but limited reuse. A high RI denotes strong scientific recognition, showing that reused datasets are also cited in academic work, whereas a low RI suggests limited citation activity, likely due to recent publication, specialized scope, or early-stage adoption within the community. Table 5 shows the top 25 lists by RES, which paints quite a

different picture to the Top 25 by total views and downloads shown in Figure 5. Thus in the context of newly added lists, the RES may help capture short-term engagement and emerging scientific recognition.

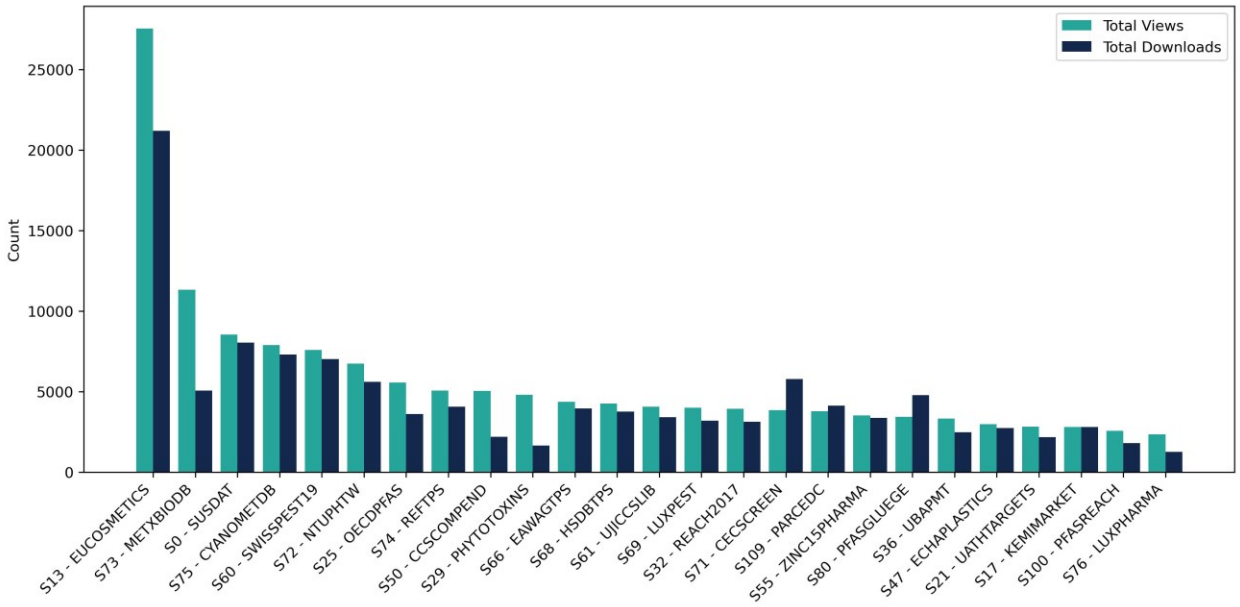

Figure 5: Total view and download statistics for the top 25 SLE lists

Sorting the SLE lists on RES reveal that 17 out of top 25 (see Table 5) are already newly added lists (above S98) suggesting that these lists are already being actively integrated into research workflows. Meanwhile, sorting the SLE lists on RI reveals S116 REFCCS at 5th position. This exemplifies early insight into the transition from reuse to citation, that can help identify lists that are beginning to gain scientific acknowledgment despite their recent release.

Table 5: Top 25 NORMAN-SLE lists based on Reuse Efficiency Score (RES)

List	Code	Total views	Total downloads	Reuse Efficiency Score (RES)	Recognition Index (RI)
FCM	S127	102	535	5.25	0
NANJINGOPDE	S131	72	272	3.78	0
FLUOROLCM	S128	74	279	3.77	0
UJIGCAPCICCS	S132	174	463	2.66	0
MTMDUST	S101	183	468	2.56	0
AQUALITY20	S107	161	395	2.45	0
PBTPFASNTA	S122	100	239	2.39	0
NDSEXTRAS	S130	27	59	2.19	0
EAWAGBBD	S121	151	286	1.89	0
STATINS	S44	437	810	1.85	0.001234
UFZWANATARG	S53	627	1141	1.82	0.000876
NEUROTOXINS	S43	537	962	1.79	0.002079

List	Code	Total views	Total downloads	Reuse Efficiency Score (RES)	Recognition Index (RI)
HDXNOEX	S42	528	925	1.75	0.002162
SINLIST	S108	327	559	1.71	0
LITCHEMPLAST	S123	130	220	1.69	0
UBADWGW	S63	487	823	1.69	0.001215
REFCCS	S116	323	540	1.67	0.003703
ANSESEDC	S99	464	774	1.67	0
PHENANTIOX	S30	550	907	1.65	0.002205
NORMANCT15	S3	945	1548	1.64	0.001938
WRIGCHRMS	S51	592	968	1.64	0.002066
EAWAGPMT	S82	474	771	1.63	0.002594
TBUTYLPHENOLS	S67	501	813	1.62	0.001230
ATHENSSUS	S8	619	1000	1.62	0.003
SURFPRO	S129	111	178	1.60	0

To summarize, the NORMAN-SLE trajectory clearly demonstrates that FAIRness directly correlates with measurable impact. The significant rise in engagement and key performance indicators is a proof that NORMAN-SLE's growth is indeed impactful.

Integration and Curation Impact:

The NORMAN-SLE contributes to PubChem through substance depositions that consolidate curated chemical content from all suspect lists. These depositions form the foundation of the NORMAN-SLE–PubChem integration and fill data gaps while serving as a crucial link between related chemical databases. As of 1 October 2025, the PubChem substance deposition included 139,400 individual substances identifiers (PubChem SIDs), corresponding to 133,573 unique compound identifiers (CIDs), as reflected in the compiled CID mappings at the top of the PubChem NORMAN-SLE Tree [80] (see also Table 3). Recently, multiple previously undefined entries were successfully registered in PubChem via the S127_LCM (4 new substances) and S129_SURFPRO (~1300 new substances!! **Emma can you please help verify the number from deposition file,I got these comparing PubChem CID changes in 2 versions**) lists, generating new PubChem Compound IDs (CIDs) further expanded it's compound coverage. [*ES: can't access deposition system atm due to shutdown, TBD*] Additionally, the integration of experimental CCS data from S116_REFCCS and S132_UJIGCAPCICCS lists expanded the collection of CCS in PubChem further, complementing related resources such as CCSbase and the BakerLab. These values are compiled together in the Aggregated CCS browser in PubChem [82].

Beyond data expansion, the period from 2022 to 2025 saw substantial quality improvements through curation. For instance, challenges arising from collaboratively generated lists led to a revision of the NORMAN-SLE workflow Standard Operating Procedure (SOP). While curating the S109 PARCEDC source data, inconsistencies were found between the submitted compound names and their associated identifiers (NORMAN IDs, CAS numbers, and SMILES). Upon thorough investigation, the issue appeared to result from misalignment or sorting errors in the names column, likely caused by multiple collaborators working on the same list. The discrepancies initially went unnoticed because the first few

rows seemed error-free. Since then, the first step in SLE curation has been updated to include adding an index number column, regardless of whether a user-provided index already exists.

Another instance was observed during the curation of the FCCMIGEX database to generate S112 FCCMIGEX. A series of incorrect CAS numbers, likely resulting from manual entry or character substitution errors, was detected. Feedback containing the proposed corrections (see Figure 6) was provided to the database team leading to updates in the official FCC database. This highlights the importance of continuous data validation and collaborative feedback mechanisms.

Index	provided CAS number	Common name	corrected CAS number	Action
867	152-05-2	CYCLODECANOL	1502-05-2	0 added
1062	175-06-5	ETHANONE, 2,2-DIHYDROXY-1-PHENYL-	1075-06-5	0 added
3540	775-01-~~0~~9	2-(3,4-DIHYDROXYPHENYL)-2-HYDROXYACETIC ACID	775-01-9	0 removed
1417	235-05-~~0~~7	DODECALACTONE	2305-05-7	0 removed

Figure 6: Snapshot of feedback on Incorrect CAS Numbers Identified During FCCMIGEX Database Curation

Future Perspective:

While Zenodo provides a reliable place to archive individual suspect lists, the increasing size and complexity of SLE data highlight the need for a more integrated, user-friendly platform to browse and explore the data. A relational database with a dedicated website could offer features that static repositories cannot, including interactive visualizations, structure-based searches, cross list browsing and overlap search download. This would make the platform more engaging and useful for a wide range of users. Hosting such a platform in an academic setting would make it easier for researchers to explore, navigate, and analyze NORMAN-SLE data. Discoverability and usability would improve with searchable metadata and summary visualizations, ensuring FAIR data have a greater impact. First scoping plans are underway, but these developments are beyond the scope of this current article.

Conclusion:

Between 2022 and 2025, the NORMAN Suspect List Exchange (NORMAN-SLE) grew steadily from 99 to 133 lists, now covering over 133,500 unique compounds contributed by >100 researchers worldwide. This steady growth reflects an active community working together to make chemical data more open, reliable, and useful. The repository now covers a broader range of chemical classes including PFAS, EDCs, FCCs, and LCMs, aligning with EU policy goals like the Green Deal and the Chemicals Strategy for Sustainability. The adoption of FAIR data principles has led to measurable impact. Views, downloads, and citations have increased significantly, showing that open and accessible data promote reuse and recognition. Integration with PubChem and continuous curation have further improved data quality, visibility and interoperability.

The NORMAN-SLE's open access model supports international efforts toward transparency in chemical risk assessment. It stands as clear evidence that open, curated, and community-driven data can transform environmental science and strengthen global collaboration.

Availability of Data and Materials

All data integrated in the NORMAN Suspect List Exchange are available from the NORMAN-SLE website (https://www.norman-network.com/nds/SLE/) and on the Zenodo NORMAN-SLE community website (https://zenodo.org/communities/norman-sle) or via the individual DOIs. The merged NORMAN SusDat collection is also available (https://www.norman-network.com/nds/susdat/). The collection on CompTox can be found under this search URL (https://comptox.epa.gov/dashboard/chemical-lists?search=NORMAN) or on the NORMAN-SLE website (https://www.norman-network.com/nds/SLE/). The NORMAN-SLE is available in PubChem (https://pubchem.ncbi.nlm.nih.gov/source/23819) and browsable as a classification tree (https://pubchem.ncbi.nlm.nih.gov/classification/#hid=101). Detailed annotation content is available in >40,000 PubChem compound records, with an overview on the Data Source page (https://pubchem.ncbi.nlm.nih.gov/source/23819). The code supporting the NORMAN-SLE (https://gitlab.com/uniluxembourg/lcsb/eci/norman-sle) and the NORMAN-SLE/PubChem integration (https://gitlab.com/uniluxembourg/lcsb/eci/pubchem) is available on GitLab.

Abbreviations

Acknowledgements

…

Funding

…

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

References

1. Feng W, Deng Y, Yang F, et al (2023) Systematic Review of Contaminants of Emerging Concern (CECs): Distribution, Risks, and Implications for Water Quality and Health. Water 15:3922. https://doi.org/10.3390/w15223922

2. Hollender J, Schymanski EL, Ahrens L, et al (2023) NORMAN guidance on suspect and non-target screening in environmental monitoring. Environ Sci Eur 35:75. https://doi.org/10.1186/s12302-023-00779-4

3. Heller S, McNaught A, Stein S, et al (2013) InChI - the worldwide chemical structure identifier standard. Journal of Cheminformatics 5:7. https://doi.org/10.1186/1758-2946-5-7

4. Heller SR, McNaught A, Pletnev I, et al (2015) InChI, the IUPAC International Chemical Identifier. Journal of Cheminformatics 7:23. https://doi.org/10.1186/s13321-015-0068-4

5. Daylight Theory: SMILES

6. CAS REGISTRY

7. Williams AJ, Grulke CM, Edwards J, et al (2017) The CompTox Chemistry Dashboard: A community data resource for environmental chemistry. Journal of Cheminformatics 9:61. https://doi.org/10.1186/s13321-017-0247-6

8. Kim S, Chen J, Cheng T, et al (2025) PubChem 2025 update. Nucleic Acids Res 53:D1516–D1525. https://doi.org/10.1093/nar/gkae1059

9. Krier J, Singh RR, Kondić T, et al (2022) Discovering pesticides and their TPs in Luxembourg waters using open cheminformatics approaches. Environment International 158:106885. https://doi.org/10.1016/j.envint.2021.106885

10. Freudenthal O, Da Silveira M, Deladiennee L (2024) Unlocking the potential of data harmonization and FAIRness in chemical risk assessment: Lessons from practice and insights for policy development. Environmental Sciences Europe 36:194. https://doi.org/10.1186/s12302-024-01022-4

11. Schymanski EL, Bolton EE (2022) FAIRifying the exposome journal: Templates for chemical structures and transformations. Exposome 2:osab006. https://doi.org/10.1093/exposome/osab006

12. Mustafa F, Lynch I, Theunis J, et al (2025) Reusing Chemical Data Across Disciplines: Initiatives and Common Challenges. Chemistry International 47:12–16. https://doi.org/10.1515/ci-2025-0203

13. Mohammed Taha H, Aalizadeh R, Alygizakis N, et al (2022) The NORMAN Suspect List Exchange (NORMAN-SLE): Facilitating European and worldwide collaboration on suspect screening in high resolution mass spectrometry. Environmental Sciences Europe 34:104. https://doi.org/10.1186/s12302-022-00680-6

14. NORMAN Association (2022) NORMAN Suspect List Exchange: Zenodo Community

15. Andres S (2022) S99 ANSESEDC List of potential endocrine disrupting compounds (EDCs) from ANSES

16. Arp HPH, Schymanski E (2025) S100 PFASREACH List of PFAS identified in REACH 2019

17. Dubocq F, Wang T (2022) S101 MTMDUST List of chemicals characterised in indoor dust samples

18. Dubocq F, Wang T (2022) Extended characterization of the indoor dust chemical composition by retrospective suspect and non-target analysis of high resolution mass spectrometric data. jeea 1:N/A–N/A. https://doi.org/10.20517/jeea.2022.23

19. Polesello S, Valsecchi S (2023) S102 PARCPFAS List of PFAS from PARC WP4

20. Schymanski E, Bolton E, Singh R, et al (2024) S103 NORMANUVCB NORMAN Dataset of Curated UVCB Mappings

21. Sims K (2023) S104 UKVETMED UK Veterinary Medicines Directorate's List

22. Assessment O of EHH (2023) S105 P65CHEM Californian Proposition 65 warning list of chemicals

23. Assoumani A (2023) S106 AQUAlity19 EU AQUAlity's water analysis list 2019

24. Assoumani A (2023) S107 AQUAlity20 EU AQUAlity's water analysis list 2020

25. Lennquist A, Arp HPH (2024) S108 SINLIST SIN (Substitute It Now) list of hazardous chemicals by ChemSec

26. Andres S, Dulio V (2024) S109 PARCEDC List of 7074 potential endocrine disrupting compounds (EDCs) by PARC T4.2

27. Derksen A (2023) S110 DUTCHUSE Dutch Prioritized Chemical Use Categories

28. Schymanski E (2023) S111 PMTPFAS Fluorine-containing Compounds in PMT Suspect Lists

29. Geueke B, Parkinson L (2025) S112 FCCMIGEX List of Migrating & Extractable Food Contact Chemicals (FCCmigex) by FPF

30. Geueke B, Groh KJ, Maffini MV, et al (2023) Systematic evidence on migrating and extractable food contact chemicals: Most chemicals detected in food contact materials are not listed for use. Critical Reviews in Food Science and Nutrition 63:9425–9435. https://doi.org/10.1080/10408398.2022.2067828

31. Meyer C, Hollender J (2025) S113 SWISSPHARMA24 2024 Swiss Pharmaceutical List with Metabolites

32. Meyer C, Stravs MA, Hollender J (2024) How Wastewater Reflects Human Metabolism–Suspect Screening of Pharmaceutical Metabolites in Wastewater Influent. Environ Sci Technol 58:9828–9839. https://doi.org/10.1021/acs.est.4c00968

33. Löffler P, Lai FY (2024) S114 SLUAMTPS Antimicrobial Transformation Products from SLU

34. Löffler P, Escher BI, Baduel C, et al (2023) Antimicrobial Transformation Products in the Aquatic Environment: Global Occurrence, Ecotoxicological Risks, and Potential of Antibiotic Resistance. Environ Sci Technol 57:9474–9494. https://doi.org/10.1021/acs.est.2c09854

35. Oswald P (2024) S115 EIEQSDWLGC EQS Directive and Watch List for GC from Environmental Institute

36. Muller H, Palm E, Schymanski E (2024) S116 REFCCS Collision Cross Section (CCS) Values from Literature

37. Phelps DW, Parkinson L, Boucher JM, et al (2024) S117 PFASFCCDB 140 PFAS from FCCdb

38. Phelps DW, Parkinson LV, Boucher JM, et al (2024) Per- and Polyfluoroalkyl Substances in Food Packaging: Migration, Toxicity, and Management Strategies. Environ Sci Technol 58:5670–5684. https://doi.org/10.1021/acs.est.3c03702

39. Phelps DW, Parkinson L, Boucher JM, et al (2024) S118 PFASFCCMIGEX 68 PFAS in Migrating & Extractable Food Contact Chemicals (FCCmigex)

40. Wagner M, Monclús L, Arp HPH, et al (2024) S119 PLASTCHEM PlastChem chemicals

41. Wagner M, Monclús L, Arp HPH, et al (2024) State of the science on plastic chemicals - Identifying and addressing chemicals and polymers of concern. Zenodo

42. Haglund P, Schymanski E (2024) S120 DUSTCT2024 Substances from Second NORMAN Collaborative Dust Trial

43. Haglund P, Alygizakis NA, Covaci A, et al (2024) Comprehensive characterization of European house dust contaminants: Concentrations and profiles, geographical variability, and implications for chemical regulation and health risk. Science of The Total Environment 957:177639. https://doi.org/10.1016/j.scitotenv.2024.177639

44. Technology SFI of AS, Palm E and (2024) S121 EAWAGBBD Curated Transformation Reactions from Eawag-BBD

45. University of Minnesota Biocatalysis/Biodegradation Database: Improving public access Nucleic Acids Research Oxford Academic

46. Han B-C, Liu J-S, Bizimana A, et al (2025) S122 PBTPFASNTA NTA prioritized PBT-PFAS compounds

47. Identifying priority PBT-like compounds from emerging PFAS by nontargeted analysis and machine learning models - PubMed

48. Helene W, Shalin A, Huang X, et al (2025) S123 LITCHEMPLAST LitChemPlast database chemicals

49. LitChemPlast: An Open Database of Chemicals Measured in Plastics Environmental Science & Technology Letters

50. Tisler S, Zweigle J, Schymanski E (2025) S124 IONICLIQUIDS Ionic Liquids Suspect List

51. Tisler S, Zweigle J, Schymanski E (2025) S125 FLUOROIONS Fluorinated Ionic Liquids Suspect List

52. Tisler S, Zweigle J, Schymanski E (2025) S126 SFCPFASFIONS PFAS and Fluorinated Ions Detected with SFC

53. Su H, Ren K, Li R, et al (2025) S127 LCM 1173 Liquid Crystal (LCM) Monomers

54. Su H, Ren K, Li R, et al (2022) Suspect Screening of Liquid Crystal Monomers (LCMs) in Sediment Using an Established Database Covering 1173 LCMs. Environ Sci Technol 56:8061–8070. https://doi.org/10.1021/acs.est.2c01130

55. Su H, Ren K, Li R, et al (2025) S128 FLUOROLCM Fluorinated Liquid Crystal (LCM) Monomers

56. Suspect Screening of Liquid Crystal Monomers (LCMs) in Sediment Using an Established Database Covering 1173 LCMs Environmental Science & Technology

57. Hödl SL, Hermans L, Dankloff PFJ, et al (2025) S129 SURFPRO Surfactant Property (SURFPRO) Database

58. Hödl SL, Hermans L, Dankloff PFJ, et al (2025) SurfPro – a curated database and predictive model of experimental properties of surfactants. Digital Discovery 4:1176–1187. https://doi.org/10.1039/D4DD00393D

59. Mohammed Taha H, Mayahi B, Palm EH, Schymanski E (2025) S130 NDSEXTRAS NORMAN Database System (NDS) Extra Entries

60. Ye L, Su G (2025) S131 NANJINGOPDE Organophosphate diester (OPDE) list from Ye & Su, Nanjing

61. Izquierdo-Sandoval D, Fabregat-Safont D, Lacalle-Bergeron L, et al (2025) S132 UJIGCAPCICCS A GC-APCI-IMS-HRMS CCS Library from Izquierdo-Sandoval et al. (2022)

62. Alygizakis N, Jonkers T (2022) S6 ITNANTIBIOTIC Antibiotic List: ITN MSCA ANSWER

63. The impact of on-site hospital wastewater treatment on the downstream communal wastewater system in terms of antibiotics and antibiotic resistance genes - ScienceDirect

64. LCSB-ECI, Krier J, Palm E, et al (2025) S68 HSDBTPS Transformation Products Extracted from HSDB Content in PubChem

65. Hazardous Substances Data Bank (HSDB) - PubChem Data Source

66. Schymanski E, Chirsir P, Baesu A, et al (2025) S74 REFTPS Transformation Products and Reactions from Literature

67. Janssen EM-L, Jones MR, Pinto E, et al (2024) S75 CyanoMetDB Comprehensive database of secondary metabolites from cyanobacteria

68. Jones MR, Pinto E, Torres MA, et al (2021) CyanoMetDB, a comprehensive public database of secondary metabolites from cyanobacteria. Water Research 196:117017. https://doi.org/10.1016/j.watres.2021.117017

69. Chirsir P, Schymanski E, Zweigle J (2025) S96 ECIPFAS Updatable List to add PFAS Structures to Public Resources from ECI (UniLu)

70. Schymanski E, Hakkinen P(Bert), Mohammed Taha H (2024) S98 TIRECHEM Tire-Related Chemicals in Environment from Literature

71. CompTox Chemicals Dashboard

72. Palm EH, Chirsir P, Krier J, et al (2023) ShinyTPs: Curating Transformation Products from Text Mining Results. Environ Sci Technol Lett 10:865–871. https://doi.org/10.1021/acs.estlett.3c00537

73. Schymanski EL, Zhang J, Thiessen PA, et al (2023) Per- and Polyfluoroalkyl Substances (PFAS) in PubChem: 7 Million and Growing. Environmental Science & Technology 57:16918–16928. https://doi.org/10.1021/acs.est.3c04855

74. Ge J, Du B, Shen M, et al (2023) A review of liquid crystal monomers: Environmental occurrence, degradation, toxicity, and human exposure of an emerging class of E-waste pollutants. Environmental Pollution 335:122267. https://doi.org/10.1016/j.envpol.2023.122267

75. Salvito D, Fernandez M, Jenner K, et al (2020) Improving the Environmental Risk Assessment of Substances of Unknown or Variable Composition, Complex Reaction Products, or Biological Materials. Environmental Toxicology and Chemistry 39:2097–2108. https://doi.org/10.1002/etc.4846

76. Kortenkamp A, Scholze M, Ermler S, et al (2022) Combined exposures to bisphenols, polychlorinated dioxins, paracetamol, and phthalates as drivers of deteriorating semen quality. Environment International 165:107322. https://doi.org/10.1016/j.envint.2022.107322

77. Groh KJ, Geueke B, Martin O, et al (2021) Overview of intentionally used food contact chemicals and their hazards. Environment International 150:106225. https://doi.org/10.1016/j.envint.2020.106225

78. The European Green Deal - European Commission

79. (2020) Chemicals strategy for sustainability: Towards a toxic free environment. Publications Office of the European Union

80. NORMAN Association, NCBI/NLM/NIH (2024) PubChem Classification Browser: NORMAN Suspect List Exchange Tree (PubChem NORMAN-SLE Tree)

81. NORMAN Association, NCBI/NLM/NIH (2022) NORMAN-SLE Data Source in PubChem

82. Zhang J, Thiessen PA, Schymanski EL, et al (2025) PubChem: Aggregated CCS Classification Tree